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1 Simply Typed Lambda Calculus

$$\dfrac{\dfrac{\dfrac{x : \mathbf{A} \in y : \mathbf{A}, x : \mathbf{A}}{y : \mathbf{A}, x : \mathbf{A} \vdash x : \mathbf{A}}\ \textsc{Ty-Var}}{y : \mathbf{A} \vdash (\lambda x : \mathbf{A}.x) : \mathbf{A} \to \mathbf{A}}\ \textsc{Ty-Abs} \quad \dfrac{y : \mathbf{A} \in y : \mathbf{A}}{y : \mathbf{A} \vdash y : \mathbf{A}}\ \textsc{Ty-Var}}{y : \mathbf{A} \vdash ((\lambda x : \mathbf{A}.x)\, y) : \mathbf{A}}\ \textsc{Ty-App}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{y : \mathbf{C} \in x : \mathbf{B}, y : \mathbf{C}, z : \mathbf{B}, w : \mathbf{A}}{x : \mathbf{B}, y : \mathbf{C}, z : \mathbf{B}, w : \mathbf{A} \vdash y : \mathbf{C}}\ \textsc{Ty-Var}}{x : \mathbf{B}, y : \mathbf{C}, z : \mathbf{B} \vdash (\lambda w : \mathbf{A}.y) : \mathbf{A} \to \mathbf{C}}\ \textsc{Ty-Abs}}{x : \mathbf{B}, y : \mathbf{C} \vdash (\lambda z : \mathbf{B}.\lambda w : \mathbf{A}.y) : \mathbf{B} \to (\mathbf{A} \to \mathbf{C})}\ \textsc{Ty-Abs} \quad \dfrac{x : \mathbf{B} \in x : \mathbf{B}, y : \mathbf{C}}{x : \mathbf{B}, y : \mathbf{C} \vdash x : \mathbf{B}}\ \textsc{Ty-Var}}{x : \mathbf{B}, y : \mathbf{C} \vdash ((\lambda z : \mathbf{B}.\lambda w : \mathbf{A}.y)\, x) : \mathbf{A} \to \mathbf{C}}\ \textsc{Ty-App}$$

There exist no context Γ or type T for the two latter terms t in the exercise. For the first, w is a free variable, and we can't deduce its type without evaluating it. For the latter: x must be of a type, that both: can be evaluated by itself being applied to AND it can be evaluated without a further parameter, as to determine its time. These can't be simultaneously true.

$$\dfrac{\dfrac{\dfrac{\dfrac{z : \mathbf{A} \in \Gamma, x : \mathbf{B}, y : \mathbf{C}, z : \mathbf{A}}{\Gamma, x : \mathbf{B}, y : \mathbf{C}, z : \mathbf{A} \vdash z : \mathbf{A}}\ \textsc{Ty-Var}}{\Gamma, x : \mathbf{B}, y : \mathbf{C} \vdash (\lambda z : \mathbf{A}.z) : \mathbf{A} \to \mathbf{A}}\ \textsc{Ty-Abs} \quad \dfrac{\lightning}{\Gamma, x : \mathbf{B}, y : \mathbf{C} \vdash (w\, x\, y) : \mathbf{A}}}{\Gamma, x : \mathbf{B}, y : \mathbf{C} \vdash ((\lambda z : \mathbf{A}.z)\, (w\, x\, y)) : \mathbf{A}}\ \textsc{Ty-App}}{\Gamma, x : \mathbf{B} \vdash \lambda y : \mathbf{C}.((\lambda z : \mathbf{A}.z)\, (w\, x\, y)) : \mathbf{C} \to \mathbf{A}}\ \textsc{Ty-Abs}}{\Gamma \vdash \lambda x : \mathbf{B}.\lambda y : \mathbf{C}.((\lambda z : \mathbf{A}.z)\, (w\, x\, y)) : \mathbf{B} \to (\mathbf{C} \to \mathbf{A})}\ \textsc{Ty-Abs}$$

$$\dfrac{\dfrac{\dfrac{\lightning}{x : \tau', y : \mathbf{A} \vdash x : (\tau \to \tau')} \quad \dfrac{x : \tau' \in x : \tau', y : \mathbf{A}}{x : \tau', y : \mathbf{A} \vdash x : \tau'}\ \textsc{Ty-Var}}{x : \tau', y : \mathbf{A} \vdash x\, x : \tau'}\ \textsc{Ty-App}}{x : \tau* \vdash \lambda y : \mathbf{A}.x\, x : \mathbf{A} \to \tau'}\ \textsc{Ty-Abs}$$

2 Extension with Numbers

TODO

3 Environments and free variables

TODO

4 Compilation to untyped lambda calculus

TODO

5 Types and Programming Languages

No questions.